Superpolynomial online mistakes for isotropic Bayesian ReLU and softplus networks

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Abstract

We construct a fixed order of distinct Boolean inputs on which exact Bayesian prediction with an ordinary one-hidden-layer ReLU network makes superpolynomially many mistakes in the strong-prior limit. The weights have independent Gaussian coordinates, with identical variances within each layer. Before observing labels, the prior is conditioned on an absolute output margin at every input; training then conditions on the output sign agreeing with every revealed Boolean label. The construction works at every sufficiently large finite hidden width. For a parameter m, the input dimension and the target's Boolean circuit size are $O(m)$. A shallow ReLU teacher has width $O(m^2)$ and coefficients bounded by that width, while $\lfloor 2^{m/3} \rfloor$ designated predictions have correct-label posterior probability tending to zero. Thus the mistake bound is $\exp(\Omega(C^{1/2}))$ in the displayed teacher width C. The same bound holds for standard softplus, with a teacher of comparable size and an existential learner-width threshold. The width threshold and calibration prefix are large, and the input order is deliberately unrepresentative of the uniform distribution. The theorem concerns cumulative online mistakes, not iid sample complexity.

The proof determines an exact weighted Fourier norm on an unequal-rail parity code, proves polynomial stability, and uses calibration to exclude all competitors outside the code. A conditional-parity argument gives a sharp atom bound with rail lengths bounded independently of the logical dimension. A transfer-matrix argument, together with sparse squarefree collisions and a Vandermonde argument, supplies exponentially many distinct completion costs. Observability outside a suitable affine set and a linear rounding estimate identify the unique cheapest completion after each history. The resulting energy gaps apply to the full Gaussian posterior, including all parameterizations of each function.

1 Model and main result

A target can have a small circuit and still be learned poorly along an unfavorable sequence of inputs. The issue for a Bayesian neural network is whether its actual posterior can repeatedly prefer a wrong completion, even when the learner is arbitrarily wider than a small teacher. We prove that this can occur with ordinary ReLU activation and isotropic Gaussian coordinates. No rule for selecting among tied minimum-energy classifiers is used. Section 7 extends the construction to the smooth activation $\log(1 + e^t)$ under the same Gaussian prior and likelihood.

The input space is $X = \{-1, 1\}^d$. Write $\bar{x} = (1, x)$ and consider

$$F_\theta(x) = \sum_{j=1}^{N} v_j \,\mathrm{ReLU}(u_j \cdot \bar{x}), \qquad \mathrm{ReLU}(t) = \max(t, 0), \quad u_j \in \mathbb{R}^{d+1}. \tag{1}$$

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The first coordinate of u_j is the weight of the constant input. There are no additional biases. Independently,

$$(u_j)_i \sim \mathcal{N}(0, N\sigma^2/d), \qquad v_j \sim \mathcal{N}(0, \sigma^2/N). \tag{2}$$

Its density is proportional to $\exp[-K_N(\theta)/(2\sigma^2)]$, where

$$K_N(\theta) = \frac{d}{N}\sum_j \|u_j\|_2^2 + N\sum_j v_j^2. \tag{3}$$

The prior is first conditioned on $|F_\theta(x)| > 1/2$ at every $x \in X$. After each prediction it is conditioned on the output sign agreeing with the revealed Boolean label. Thus a history $H \subseteq X$ with target $f : X \to \{-1, 1\}$ conditions the Gaussian law on

$$|F_\theta(x)| > 1/2 \quad (x \in X), \qquad f(x)F_\theta(x) > 1/2 \quad (x \in H). \tag{4}$$

Thus training conditions on signs with the stated margin, without fixing the real-valued outputs to the label values. The strong-prior limit is $\sigma \downarrow 0$ at each fixed d, N.

For the ReLU model, the conclusion also holds for any two fixed positive layer variances $\alpha\sigma^2$ and $\beta\sigma^2$ in place of (2). Positive homogeneity gives the lower bound $2V(F)/\sqrt{\alpha\beta}$, where V is the variation norm defined below. Balancing any finite representation of cost B attains energy $2B/\sqrt{\alpha\beta}$. The constructed minimizers attain this bound at the widths used below, so their strict energy order is unchanged. In particular, writing $t = \sqrt{N}\,\sigma$ turns (2) exactly into first-layer variance t^2/d and readout variance t^2/N^2. At each fixed width the limits $t \downarrow 0$ and $\sigma \downarrow 0$ are the same.

Theorem 1 (A fixed sequence with many confident errors)**.** *For every sufficiently large integer m, put*

$$R = 2^{1024}, \qquad \ell = 2m - 1, \qquad n_0 = \ell R, \qquad n_{\max} = n_0 + 6m - 2,$$
$$d = 2n_0 + 6m - 2, \qquad N_0 = 2^{m+n_{\max}+1}, \qquad q_m = \lceil m/64 \rceil.$$

There are a Boolean target f_m on d bits, a fixed permutation of its input cube, and

$$T_m = \lfloor 2^{m/3} \rfloor$$

designated rounds such that, for every finite $N \geq N_0$, the posterior probability of the correct label at each designated round tends to zero as $\sigma \downarrow 0$.

The target has a bounded-fan-in Boolean circuit with $O(d) = O(m)$ gates and a one-hidden-layer ReLU implementation of width at most

$$C_m = (m + 1)(n_0 + 1) + 8m - 4 = \Theta(m^2),$$

output margin one, and every coefficient bounded by C_m. The same input permutation works for every $N \geq N_0$.

For every $\eta > 0$ and fixed such m, N, all sufficiently small positive σ therefore have these properties:

1. *Fresh posterior sampling makes at least T_m mistakes with probability at least $1 - \eta$.*
2. *Posterior-label majority makes a mistake at every designated round.*
3. *A prediction rule requiring correct-label posterior probability strictly greater than $2/3$ fails at every designated round.*

The width threshold is an explicit sufficient bound. Here $d = \Theta(m)$, $C_m = \Theta(m^2)$, and $q_m = \lceil m/64 \rceil$, so the theorem gives $\exp(\Omega(C_m^{1/2}))$ cumulative mistakes. The fixed rail length $R = 2^{1024}$ is deliberately generous and is not optimized. In terms of the displayed bounded-fan-in Boolean circuit, whose gate count is $S_m = O(m)$, the mistake bound is $\exp(\Omega(S_m))$. In terms of the displayed one-hidden-layer neural teacher, it is $\exp(\Omega(\sqrt{C_m}))$ in its width C_m. The width quantifier covers every fixed $N \geq N_0$; it does not claim the result for every width merely exceeding the teacher width. The calibration prefix reveals all inputs outside a small union of logical fibers. This is compatible with good iid learning guarantees.

2 Variation norm and parity codes

All Fourier coefficients and L^2 norms use the uniform physical cube. For $A \subseteq [d]$, write $\chi_A(x) = \prod_{i \in A} x_i$ and $\widehat{F}(A) = \mathbb{E}[\chi_A F]$. Define

$$V(F) = \inf \left\{ \sum_j |c_j| \|a_j\|_2 : F(x) = \sum_j c_j \, \mathrm{ReLU}(a_j \cdot \bar{x}) \text{ on } X \right\}.$$

The same arguments apply to the cost of a specified finite representation, without taking this infimum. The arithmetic–geometric mean inequality gives

$$K_N(\theta) \geq 2\sqrt{d} \sum_j |v_j| \|u_j\|_2 \geq 2\sqrt{d}\, V(F_\theta). \tag{5}$$

Conversely any finite representation can be balanced neuron by neuron so that its energy equals twice $\sqrt{d}$ times its variation cost. Unused neurons may be set to zero.

For positive even n, define

$$\kappa_n = \frac{\binom{n-2}{(n-2)/2}}{2^{n-1}}, \qquad a(n) = \frac{\sqrt{n}}{\kappa_n}. \tag{6}$$

The alternating binomial identity

$$2^{-n} \sum_{k=0}^{n} (-1)^k \binom{n}{k} \max(n - 2k, 0) = (-1)^{n/2+1} \kappa_n$$

shows that a balanced unit ridge on n coordinates attains absolute parity correlation $\kappa_n/\sqrt{n}$. Signed averaging over all input-coordinate flips represents χ_A with variation cost $a(|A|)$, using at most $2^{|A|}$ neurons.

2.1 The full rail-code atom theorem

For this theorem only, let $p \geq 2$ denote a number of disjoint rail pairs (B_i^0, B_i^1). Their full transversals are $\widetilde{A}_t = \bigcup_i B_i^{t_i}$, indexed by $t \in \mathbb{F}_2^p$. Let $\mathcal{D}_{\text{full}}$ be the span of characters contained in at least one such transversal, and let $\mathcal{A}_{\text{full}}$ be the union of their balanced parameter rays, including zero and all coordinate sign choices. Each nonzero ray has zero bias, uses one whole rail per pair, and has equal absolute coordinate weights.

Theorem 2 (Exact unequal-rail norm and stability)**.** *Suppose all rail sizes are even and belong to* $[R, R + \Delta]$*, where* R, Δ *are positive even integers,* $\Delta \geq 2$*, and*

$$R \geq 6\Delta. \tag{7}$$

For every affine ReLU atom $h_u(x) = \mathrm{ReLU}(u \cdot \bar{x})$,

$$Q_{\mathrm{full}}(h_u) := \sum_t a(|\widetilde{A}_t|)|\widehat{h}_u(\widetilde{A}_t)| \le \|u\|_2. \tag{8}$$

Nonzero equality holds exactly on $\mathcal{A}_{\mathrm{full}}$. *Consequently*

$$V\left(\sum_t c_t \chi_{\widetilde{A}_t}\right) = \sum_t a(|\widetilde{A}_t|)|c_t|. \tag{9}$$

The parameter and function stability estimates are

$$\mathrm{dist}(u, \mathcal{A}_{\mathrm{full}})^2 \le 2^9 R^2 \|u\|_2 (\|u\|_2 - Q_{\mathrm{full}}(h_u)), \tag{10}$$

$$\|h_u - P_{\mathcal{D}_{\mathrm{full}}} h_u\|_2^2 \le 2^9 R^2 \|u\|_2 (\|u\|_2 - Q_{\mathrm{full}}(h_u)). \tag{11}$$

A complete proof appears in Appendix A. The exact constant in the norm and the polynomial stability bound will both be used.

3 A path-based subcode with small physical dimension

Let $m \ge 2$ and $\ell = 2m - 1$. The rail pairs are indexed by the columns

$$a_j \in \{e_1, \dots, e_m\} \cup \{e_1 + e_2, e_2 + e_3, \dots, e_{m-1} + e_m\} \subseteq \mathbb{F}_2^m.$$

The first m columns are called loops and the others edges. Every short rail has the fixed even size $R = 2^{1024}$. The long rail has size $R + 4$ at a loop and $R + 2$ at an edge. Thus Theorem 2 applies with $p = \ell$ and $\Delta = 4$.

Retain only the transversals

$$A_s = \bigcup_{j=1}^{\ell} B_j^{a_j \cdot s} \quad (s \in G := \mathbb{F}_2^m), \qquad b = \chi_{A_0}.$$

If $Ds = (s_i + s_{i+1})_{i=1}^{m-1}$ is the path difference map, then

$$n_s = |A_s| = n_0 + 4|s| + 2|Ds|, \qquad n_0 = \ell R, \qquad n_s \le n_{\max} = n_0 + 6m - 2. \tag{12}$$

The total physical dimension is $d = 2n_0 + 6m - 2$. Let $\xi_j \in \mathbb{F}_2$ encode the parity of the whole jth rail pair, and define the logical syndrome

$$(-1)^{\xi_j} = \chi_{B_j^0 \cup B_j^1}(x), \qquad z = \sum_j a_j \xi_j.$$

The vector z is uniform because the loop columns span G, and b is independent of z. Moreover $\chi_{A_s} = b(-1)^{s \cdot z}$. Let $\mathcal{C}$ be the span of these retained characters and $\mathcal{D}$ their downward Fourier closure. In particular, an omitted complete transversal does not belong to $\mathcal{D}$. The code projection is

$$P_{\mathcal{C}} F = b g(z), \qquad g(z) = \mathbb{E}[bF \mid z]. \tag{13}$$

Put $w_s = a(n_s)$ and $Q(F) = \sum_s w_s |\widehat{F}(A_s)|$. Restricting the full atom inequality and using signed parity averaging gives

$$Q(h_u) \le \|u\|_2, \qquad V\left(\sum_s c_s \chi_{A_s}\right) = \sum_s w_s |c_s|. \tag{14}$$

Lemma 3 (Stability after restricting the transversals)**.** *Let $\mathcal{A}$ be the retained balanced parameter rays. Then*

$$\operatorname{dist}(u,\mathcal{A})^2 \le K_c\|u\|_2(\|u\|_2 - Q(h_u)), \qquad K_c = 2^{16}\ell^2R^4.$$

The same bound holds for $\|h_u - P_{\mathcal{D}}h_u\|_2^2$.

Proof. Write $\varepsilon = \|u\|_2 - Q(h_u)$ and $S_0 = 2^9R^2$. A closest full-code ray u_* satisfies $\|u-u_*\|_2^2 \le S_0\|u\|_2\varepsilon$. If its transversal is retained, use u_*. Otherwise, for its omitted transversal A_t and weight $w_t = a(|A_t|)$,

$$w_t|\widehat{h}_u(A_t)| \le Q_{\text{full}}(h_u) - Q(h_u) \le \varepsilon, \qquad w_t|\widehat{h}_{u_*}(A_t)| = \|u_*\|_2.$$

ReLU is 1-Lipschitz and $\mathbb{E}[\bar{x}\bar{x}^\top] = I$, so

$$\|u\|_2 \le \varepsilon + (1+w_{\max})\|u-u_*\|_2, \qquad w_{\max} = \max_t a(|A_t|).$$

Since $\varepsilon \le \|u\|_2$, squaring gives

$$\|u\|_2^2 \le [2(1+w_{\max})^2S_0 + 2]\|u\|_2\varepsilon.$$

Use the zero ray in this case. The elementary bounds $n \le a(n) \le 3n$ give $w_{\max} \le 3\ell(R+4) \le 6\ell R$, so the bracket is at most K_c. For the lower bound, use $a(2) \ge 2$ and $a(n+2)/a(n) = \sqrt{n(n+2)}/(n-1) \ge (n+2)/n$. For the upper bound, induction gives $\binom{2k}{k}/4^k \ge 1/(2\sqrt{k})$ for $k \ge 1$, because $(2k+1)^2 \ge 4k(k+1)$. Substituting $k = (n-2)/2$ gives $a(n) \le 2\sqrt{2n(n-2)} < 3n$ for $n \ge 4$; check $n = 2$ directly. Finally, the function generated by a retained ray belongs to $\mathcal{D}$, and the same Lipschitz estimate proves function stability. □

For a representation of variation cost B and any physical code dual Λ with $|\widehat{\Lambda}(A_s)| \le w_s$, write $D = B - \langle \Lambda, F\rangle$. Weighted Cauchy–Schwarz gives

$$\|F - P_{\mathcal{D}}F\|_2^2 \le K_cBD. \tag{15}$$

Indeed the weighted sum of atom deficits is at most D. The statement also holds for $B = V(F)$ by approximation.

Lemma 4 (Observability outside independent syndrome sets)**.** *If $U \subseteq G$ satisfies $U \cap (U + a_j) = \varnothing$ for every column a_j, then every $A \in \mathcal{D} \cap \mathcal{C}^\perp$ obeys*

$$\|A\|_2^2 \le 2\mathbb{E}[\mathbf{1}_{z\notin U}A^2]. \tag{16}$$

Proof. Each Fourier support in A is a proper subset of a retained transversal. Assign it to the first rail pair j at which its restriction is neither complete rail, and write $A = \sum_j A_j$. Conditioned on z, two physical characters have nonzero covariance only if their symmetric difference consists of full rail pairs with incidence vector $(a_i \cdot s)_i$ for some $s \in G$. At a defective pair, a proper subset of one rail cannot be complemented within the full pair while remaining a subset of a single rail. Thus the two restrictions must agree there. Comparing first defective pairs shows that A_j, A_k are conditionally orthogonal for $j \ne k$. Every Fourier mode s of $\ell_j(z) = \mathbb{E}[A_j^2 \mid z]$ has $a_j \cdot s = 0$, so $\ell_j(z + a_j) = \ell_j(z)$. The set U contains at most one endpoint of each a_j-pair. Hence $\mathbb{E}[\mathbf{1}_U\ell_j] \le \|A_j\|_2^2/2$. Summing proves the claim. □

4 Exponentially many distinct completion costs

Write $M = 2^m$ and $\widehat{w}(v) = M^{-1}\sum_s (-1)^{s\cdot v} w_s$. Appendix B proves the exact representation

$$a(r) = \sqrt{2\pi}(r - 3/4) - 2\sqrt{2\pi}\int_0^\infty e^{-rt/2} h(t)\, dt, \qquad 0 < h(t) \le 3. \tag{17}$$

Lemma 5 (Walsh signs and a high-weight bound)**.** *Every nonconstant $\widehat{w}(v)$ is strictly negative. For $|v| \ge q \ge 3$, with $k = \lceil q/2 \rceil$,*

$$|\widehat{w}(v)| \le \frac{31}{\ell R}\left(\frac{8}{R}\right)^k. \tag{18}$$

Proof. Write $n_s = n_0 + \sum_j \delta_j (a_j \cdot s)$, where the Boolean linear form is interpreted as zero or one and $\delta_j \in \{2, 4\}$. Each factor $e^{-t\delta_j (a_j\cdot s)}$ has Fourier coefficients $(1 + e^{-t\delta_j})/2$ at zero and $(1 - e^{-t\delta_j})/2$ at a_j. Therefore the Fourier coefficients of $e^{-t(n_s - n_0)}$ form the probability law of a sum of independent column-valued Bernoulli variables. Every syndrome has positive probability for $t > 0$, since the loop columns alone span G. The affine part of (17) has nonpositive nonconstant coefficients, while its negative integral has strictly negative ones. This proves the sign assertion.

Each column has Hamming weight at most two. A column subset summing to v therefore has at least k members. Each Bernoulli success probability is at most $\delta_j t/2 \le 2t$, so the union bound over k successes bounds the v-coefficient by $\binom{\ell}{k}(2t)^k$. When $|v| \ge 3$, the affine part has zero coefficient there. Rescale the integral in (17) by $t/2 \mapsto t$. Its density is bounded by $12\sqrt{2\pi}$, whence

$$|\widehat{w}(v)| \le 12\sqrt{2\pi}\binom{\ell}{k} 2^k \int_0^\infty e^{-n_0 t} t^k\, dt \le \frac{31}{n_0}\left(\frac{2\ell}{n_0}\right)^k.$$

This implies the stated, weaker bound. □

Lemma 6 (Distinct scalar levels from a path transfer matrix)**.** *The degree polynomials*

$$P_v(X) = \sum_{s\in G} (-1)^{s\cdot v} X^{(n_s - n_0)/2}$$

have at least $2^{m/2}$ distinct values as v varies over G.

Proof. Let B be the vertex-edge incidence matrix of the path and put $a = \tanh(2t)$, $e = \tanh t$. Expanding the Fourier factors just used gives

$$\widehat{e^{-tn_s}}(v) = e^{-tn_0}\left(\frac{1 + e^{-4t}}{2}\right)^m \left(\frac{1 + e^{-2t}}{2}\right)^{m-1} Z_v(a, e),$$

$$Z_v(a, e) = \sum_{y\in \mathbb{F}_2^{m-1}} e^{|y|} a^{|v + By|}.$$

Define

$$M_0 = \begin{pmatrix} 1 & ae \\ a & e \end{pmatrix}, \qquad M_1 = \begin{pmatrix} a & e \\ 1 & ae \end{pmatrix}, \qquad c_0 = \begin{pmatrix} 1 \\ a \end{pmatrix}, \qquad c_1 = \begin{pmatrix} a \\ 1 \end{pmatrix}.$$

Successively summing the edge bits gives

$$\begin{pmatrix} Z_v \\ Z_{v+e_1} \end{pmatrix} = M_{v_1} \cdots M_{v_{m-1}} c_{v_m}.$$

Choose $t = \operatorname{arctanh}(1/2)$, so $e = 1/2$ and $a = 4/5$. On the interval $I = [4/5, 5/4]$, the ratio map of M_0 is

$$f(r) = \frac{r + ae}{ar + e}, \qquad f(I) = [20/19, 11/10] \subset (1, 5/4].$$

The ratio map of M_1 is $1/f(r)$ and lies below one in I. Both are invertible because $\det M_0 = e(1 - a^2) > 0$. The terminal ratios are $1/a$ and a. Thus Z_v/Z_{v+e_1} determines v: the side of one identifies the first bit, inversion removes that matrix, and iteration reaches the distinct terminal ratios. There are 2^m distinct ordered pairs (Z_v, Z_{v+e_1}). If there are L scalar values among all Z_v, there are at most L^2 ordered pairs, so $L \geq 2^{m/2}$.

Distinct transfer values imply distinct polynomials P_v by taking $X = e^{-2t}$ in their defining formula. This proves the claim. □

4.1 Cost separation with constant rail size

Fix the even constant $R = 2^{1024}$ and put

$$\ell = 2m - 1, \qquad n_0 = \ell R, \qquad n_s = n_0 + 4|s| + 2|Ds|.$$

The possible degrees are among

$$n_j = n_0 + 2j, \qquad 0 \leq j \leq 3m - 1.$$

Call an index j *bad* if n_j shares its squarefree part with some other $n_{j'}$ in this interval, and let B_m be the number of bad indices.

Lemma 7 (Few degree positions share a squarefree part)**.** *For $m \geq 2$,*

$$B_m \leq \frac{24m}{R} + 2\sqrt{Rm}.$$

In particular, for $m \geq 2^{20}R$, if $q = \lceil m/64 \rceil$ and $k = \lceil q/2 \rceil$, then $B_m < k$.

Proof. Suppose $ru^2 < rv^2$ are two integers in $[n_0, n_0 + 6m - 2]$ with the same squarefree part r. Their difference is at least $r(2u + 1) > 2ru = 2ru^2/u \geq 2n_0/u$. Consequently

$$u > \frac{2n_0}{6m - 2} \geq R/2.$$

The larger multiplier v satisfies the same lower bound. Thus every bad degree is divisible by h^2 for some integer $h \geq R/2$, with $h \leq \sqrt{n_0 + 6m - 2}$. Counting all integers in the interval, rather than only its even ones, gives the upper bound

$$\begin{aligned} B_m &\leq \sum_{h=R/2}^{\lfloor \sqrt{n_0+6m-2} \rfloor} \left(\frac{6m - 2}{h^2} + 1 \right) \\ &\leq \frac{6m}{R/2 - 1} + \sqrt{n_0 + 6m - 2} \leq \frac{24m}{R} + 2\sqrt{Rm}. \end{aligned}$$

We used $R \geq 4$ and $n_0 + 6m - 2 \leq 4Rm$. For $m \geq 2^{20}R$, the second term is at most $m/512$; the first is strictly less than $m/512$. Hence $B_m < m/256 < m/128 \leq k$. □

For $v \in \mathbb{F}_2^m$, define the signed degree polynomial

$$P_v(X) = \sum_{s \in \mathbb{F}_2^m} (-1)^{s \cdot v} X^{(n_s - n_0)/2}.$$

Lemma 6 establishes at least $2^{m/2}$ distinct polynomials independently of R.

Lemma 8 (A high-weight syndrome gives a high-order zero)**.** *If $|v| \geq q$, then $(1-X)^k$ divides $P_v(X)$, where $k = \lceil q/2 \rceil$.*

Proof. Index the columns by the loops $a_j = e_i$ and the path edges $a_j = e_i + e_{i+1}$. Let $r_j = 2$ on a loop and $r_j = 1$ on an edge. Then

$$X^{(n_s - n_0)/2} = \prod_j X^{r_j (a_j \cdot s)},$$

where each Boolean linear form is interpreted as zero or one. Expand

$$X^{r_j (a_j \cdot s)} = \frac{1 + X^{r_j}}{2} + \frac{1 - X^{r_j}}{2} (-1)^{a_j \cdot s}.$$

After Fourier averaging, only subsets J of columns with $\sum_{j \in J} a_j = v$ remain. Every column has Hamming weight at most two, so every such J has $|J| \geq \lceil |v|/2 \rceil \geq k$. Each corresponding term contains at least k factors divisible by $1 - X$. This proves the asserted polynomial divisibility. □

Lemma 9 (High-weight polynomial distinctions survive in the costs)**.** *Let $m \geq 2^{20} R$, $q = \lceil m/64 \rceil$, and $w_s = a(n_s)$, with $a(n) = \sqrt{n}/\kappa_n$ the ReLU parity cost. For $|v|, |v'| \geq q$,*

$$\widehat{w}(v) = \widehat{w}(v') \quad \Longleftrightarrow \quad P_v = P_{v'}.$$

Proof. The reverse implication is immediate. For the forward implication, each $a(n_j)$ is a nonzero rational multiple of $\sqrt{n_j}$. Distinct squarefree radicals are linearly independent over $\mathbb{Q}$: sign-changing automorphisms of the field generated by their prime square roots separate the corresponding characters, and averaging against each character isolates its coefficient. Thus equality of the two Walsh costs forces equality of the signed counts at every degree whose squarefree class occurs only once. The difference $P = P_v - P_{v'}$ is therefore supported on at most B_m monomials, all at bad degree positions.

By Lemma 8, P is divisible by $(1-X)^k$, and Lemma 7 gives $B_m < k$. A nonzero polynomial with b nonzero monomials cannot have a zero of multiplicity b at $X = 1$. Indeed, write it as $\sum_{i=1}^{b} c_i X^{e_i}$ with distinct exponents. Applying $(X\, d/dX)^j$ at one for $0 \leq j < b$ gives $\sum_i c_i e_i^j = 0$ if that multiplicity is at least b. The resulting Vandermonde matrix is invertible, a contradiction. Hence $P = 0$. □

Lemma 10 (An observable affine set retaining many levels)**.** *For $q = q_m$ and all sufficiently large m, there is an affine coset $U = v_0 + H$ and a subset $\mathcal{V} \subseteq U$ of size T_m such that every $v \in U$ has $|v| \geq q$, the costs indexed by $\mathcal{V}$ are pairwise distinct, and $U \cap (U + a_j) = \varnothing$ for every column. The codimension of H is between q and $q + 2$.*

Proof. Take one representative of each distinct polynomial in Lemma 6. Discard those of weight below q, at most $\sum_{j<q} \binom{m}{j}$ points. Lemma 9 makes the remaining costs pairwise distinct. Each remaining point contains a q-subset J of its support. Pigeonhole over $\binom{m}{q}$ such subsets and the four

values of the two parities with normal vectors $\mathbf{1} = (1, \ldots, 1)$ and $a_{\text{alt}} = (1, 0, 1, 0, \ldots)$. Some affine set

$$U = \{z : z_j = 1 \ (j \in J), \quad \mathbf{1} \cdot z = \alpha, \quad a_{\text{alt}} \cdot z = \beta\} \tag{19}$$

contains at least T_m of these representatives. Select that many as $\mathcal{V}$. Ties in all selections may be settled lexicographically. The set is nonempty and excludes zero. Its defining normals have rank between q and $q+2$. A loop translation changes the all-ones parity; an edge translation changes the alternating parity. Thus U is independent in every column direction.

For $q = \lceil m/64 \rceil$ and sufficiently large m,

$$\binom{m}{q}, \quad \sum_{j<q} \binom{m}{j} \le (em/q)^q \le 2^{m/8}.$$

For the binomial sum, use $t = q/m \le 1$ and $t^q \sum_{j \le q} \binom{m}{j} \le (1+t)^m \le e^q$. Also $q \log_2(em/q) \le (m/64+1)(6 + \log_2 e) < m/8$ for sufficiently large m. Thus the affine pigeonhole step retains at least $2^{3m/8-3} > T_m$ representatives for large m; select exactly T_m. □

5 Calibration that forces a single exceptional fiber

Put $M = 2^m$ and use $U = v_0 + H$ and $\mathcal{V}$ from Lemma 10. Write $K = |U|$, $r = \operatorname{codim} H$, $J_0 = H + \operatorname{span}(v_0)$, and $E = H^\perp \setminus J_0^\perp$. Then $|E| = 2^{r-1} \ge 2$, and every character in E equals -1 on U.

The target logical label is

$$t(z) = \begin{cases} +1, & z = 0, \\ -1, & z \ne 0, \end{cases} \qquad f(x) = b(x)t(z(x)).$$

Initially reveal every physical input whose logical coordinate lies outside $\mathcal{V}$, including all fibers in $U \setminus \mathcal{V}$. Subsequently whole fibers of $\mathcal{V}$ will be revealed in a specified order. Let $U_{\text{live}} \subseteq \mathcal{V}$ be the nonempty set of fibers not yet revealed. All known labels on a removed fiber are the negative logical label. Write $\mathcal{K} = \{x : z(x) \notin U\}$ for the fixed calibration region. Extra revealed fibers impose additional constraints but are not included in averages with subscript cal.

For $v \in U$, define the star score

$$g_v(z) = -\tfrac{1}{2} + \mathbf{1}_{z=0} + \mathbf{1}_{z=v}, \qquad F_v(x) = b(x)g_v(z(x)).$$

It agrees with the target on every fiber except v, and has absolute margin $1/2$ everywhere. Its code coefficients have negative sign at $s = 0$, are nonnegative elsewhere, and vanish on E.

Define the code dual through its logical Fourier coefficients:

$$\widehat{\lambda}_s = w_s - 2w_0 \mathbf{1}_{s=0} - p_E \mathbf{1}_{s \in E}, \qquad p_E = \frac{2w_0}{|E|}. \tag{20}$$

Its physical version is $b(x)\lambda(z(x))$. Since $w_s \ge w_0$, $|\widehat{\lambda}_s| \le w_s$, so it is a unit variation-norm dual. Fourier inversion gives

$$\lambda(z) = M\widehat{w}(z) - 2w_0 - 2w_0 \mathbf{1}_H(z) + 2w_0 \mathbf{1}_U(z). \tag{21}$$

In particular $\lambda(v) = M\widehat{w}(v)$ on U, while on the fixed calibration region its sign agrees with t and its magnitude is at least $2w_0$.

The stars saturate the dual and have exact costs

$$V(F_v) = C_v = C_* + \widehat{w}(v), \qquad C_* = \frac{1}{M}\sum_s w_s + \frac{w_0}{2} - \frac{4w_0}{M}. \tag{22}$$

For $v \in \mathcal{V}$, Lemma 10 makes these costs pairwise distinct. Only these representatives can remain live.

5.1 A linear rounding estimate

Consider any finite ReLU score F with absolute margin at least $1/2$ on the full cube and correct margins on all known fibers. Let B be its variation cost, or its variation norm. We only need consider $B \le \max_v C_v$, since the remaining stars are competitors. Write its code projection as bg, and set

$$p_v = g(v) + \tfrac{1}{2} \quad (v \in U), \qquad D = B - \langle b\lambda, F\rangle,$$
$$S = \mathbb{E}_{\mathrm{cal}}[|\lambda(z)|(fF - \tfrac{1}{2})].$$

The expectation with subscript "cal" is the original normalized physical expectation with the indicator of $\mathcal{K}$. Both D and S are nonnegative. A direct decomposition gives

$$B = C_* + D + S + \sum_{v\in U} \widehat{w}(v)p_v. \tag{23}$$

Lemma 11 (Physical variance control)**.** *For these scores,*

$$L := \|F - P_{\mathcal{C}}F\|_2^2 \le C_{\mathrm{phys}}(D + S), \qquad C_{\mathrm{phys}} = 2^{22}\ell^3R^5. \tag{24}$$

Proof. On a fiber in the fixed calibration region put $e = fF - 1/2 \ge 0$. The signed conditional expectation (13) gives

$$\mathbb{E}_{\mathrm{cal}}|F - P_{\mathcal{C}}F| \le 2\mathbb{E}_{\mathrm{cal}}e \le S/w_0.$$

A variation-cost-B representation satisfies $\|F\|_\infty \le \sqrt{d+1}B$, and $P_{\mathcal{C}}$ is an L^∞ contraction. Hence the observed squared residual is at most $4\sqrt{d+1}B\mathbb{E}_{\mathrm{cal}}e$. Write $E_0 = F - P_{\mathcal{D}}F$ and $A = P_{\mathcal{D}}F - P_{\mathcal{C}}F$. Lemma 4, $(u-v)^2 \le 2u^2 + 2v^2$, and (15) give

$$\|A\|_2^2 \le 16\sqrt{d+1}B\mathbb{E}_{\mathrm{cal}}e + 4\|E_0\|_2^2, \qquad \|E_0\|_2^2 \le K_cBD.$$

The two residuals are orthogonal, so

$$L \le B\left[\frac{8\sqrt{d+1}}{w_0}S + 5K_cD\right].$$

We may use $B \le 5\ell(R+4) \le 10\ell R$, $w_0 \ge \ell R$, and $d+1 \le 4\ell R$. The coefficients are bounded by $160\sqrt{\ell R}$ and $50\cdot 2^{16}\ell^3R^5$, respectively. Both are at most $C_{\mathrm{phys}} = 2^{22}\ell^3R^5$. □

Lemma 12 (Rounding to a live star)**.** *For some $v \in U_{\mathrm{live}}$,*

$$\|p - e_v\|_1 \le B_{\mathrm{round}}(D + S), \qquad B_{\mathrm{round}} = 2^{25}2^m\ell^3R^5. \tag{25}$$

Here e_v is the point mass at v, indexed by all of U.

Proof. Put $d_z = g(z) - (-1/2 + \mathbf{1}_{z=0})$ outside U. Calibration and (21) imply

$$\sum_{z \notin U} |d_z| \le \frac{MS}{2w_0}.$$

The Fourier norm inequality and dual slack give

$$\sum_{s \in E} |\widehat{g}(s)| \le D/p_E, \qquad 2w_s(-\widehat{g}(s))_+ \le D \quad (s \notin E \cup \{0\}).$$

Set $\eta = M(D+S)/(2w_0)$. For every $s \in E$,

$$M\widehat{g}(s) = 1 - \sum_{v \in U} p_v + \sum_{z \notin U} d_z (-1)^{s \cdot z}.$$

Average the absolute-value bound over E and use $|E|p_E = 2w_0$. This gives

$$\left| \sum_{v \in U} p_v - 1 \right| \le \eta.$$

For any nontrivial character of H, choose two extensions to G having opposite signs on v_0. They differ by an element of E and both lie outside $H^\perp$. Applying the negative-coefficient inequality to these two extensions bounds the absolute value of the unnormalized Fourier transform of p by $1 + \eta$. The zero frequency obeys the same bound. Parseval on the affine cube U gives

$$\sum_{v \in U} p_v^2 \le (1+\eta)^2. \tag{26}$$

Every value of the signed physical score bF has absolute value at least $1/2$. Its conditional mean on a fiber v is $p_v - 1/2$. Therefore

$$V_{\mathrm{frac}} := \sum_{0 < p_v < 1} p_v(1 - p_v) \le ML.$$

Suppose first $\eta \le 1$. Then $\sum p_v^2 - \sum p_v \le 4\eta$, so

$$\sum_{p_v \notin [0,1]} (p_v^2 - p_v) \le 4\eta + V_{\mathrm{frac}}.$$

Round every coordinate to its nearest $b_v \in \{0, 1\}$. The distance is at most $p_v^2 - p_v$ outside $[0, 1]$ and at most $2p_v(1 - p_v)$ inside. Consequently

$$\|p - b\|_1 \le 4\eta + 3V_{\mathrm{frac}}, \qquad \left| \sum b_v - 1 \right| \le 5\eta + 3V_{\mathrm{frac}}.$$

A revealed whole fiber has $p_v \le 0$, hence $b_v = 0$. If b is nonzero, choose a live v with $b_v = 1$; if $b = 0$, choose any live v. In both cases $\|b - e_v\|_1 = |\sum b_v - 1|$, so

$$\|p - e_v\|_1 \le 9\eta + 6V_{\mathrm{frac}}.$$

For $\eta > 1$, use the logical Fourier algebra norm $\|h\|_{\mathrm{A}} = \sum_s |\widehat{h}(s)|$. Multiplication is submultiplicative for this norm, and $\|\mathbf{1}_U\|_{\mathrm{A}} = 1$. Extend p by zero outside U, obtaining $\widetilde{p} = (g + 1/2)\mathbf{1}_U$. The atom norm bound gives

$$\|\widetilde{p}\|_{\mathrm{A}} \le \|g\|_{\mathrm{A}} + \tfrac{1}{2} \le B/w_0 + \tfrac{1}{2} \le 10.5.$$

Let $P_h = \sum_{u \in H} p_{v_0+u} h(u)$ for characters h of H. Each such character has M/K extensions to G, whose coefficients of $\widetilde{p}$ have absolute value $|P_h|/M$. Consequently

$$\frac{1}{K}\sum_h |P_h| = \|\widetilde{p}\|_{\mathrm{A}} \le 10.5, \qquad \sum_v p_v^2 = \frac{1}{K}\sum_h |P_h|^2 \le 10.5(1+\eta) \le 21\eta.$$

Together with $\sum p_v \ge 1-\eta$, this yields $\sum(p_v^2 - p_v) \le 22\eta$. The same coordinate rounding now gives

$$\|p - b\|_1 \le 22\eta + 3V_{\mathrm{frac}}, \qquad \left|\sum b_v - 1\right| \le 23\eta + 3V_{\mathrm{frac}}.$$

Choosing a live point as above proves $\|p - e_v\|_1 \le 45\eta + 6V_{\mathrm{frac}}$. Thus in all cases

$$\mathrm{dist}_1(p, \{e_v : v \in U_{\mathrm{live}}\}) \le M\left(\frac{45}{2w_0} + 6C_{\mathrm{phys}}\right)(D+S).$$

Using $w_0 \ge \ell R$ and Lemma 11 bounds this constant by $2^{25} M \ell^3 R^5 = B_{\mathrm{round}}$. □

Proposition 13 (Every minimum-cost completion is the cheapest star)**.** *At every stage with $U_{\mathrm{live}} \neq \varnothing$, the unique minimum-variation-cost score consistent with all known whole fibers and the closed full-cube margins is F_v, where v is the cheapest live fiber in the order (22).*

Proof. Let $W = \max_{v \in U} |\widehat{w}(v)|$. For a score no more expensive than a live star, choose the live vertex furnished by Lemma 12. Equations (23) and (25) imply

$$B \ge C_v + (1 - W B_{\mathrm{round}})(D+S).$$

The parameter choice makes $W B_{\mathrm{round}} < 1/2$. Indeed Lemma 5, $R = 2^{1024}$, and $k = \lceil q_m/2 \rceil \ge m/128$ give

$$\log_2(W B_{\mathrm{round}}) < m + 4126 + 2\log_2 \ell - 1021k < -2$$

for all sufficiently large m.

A cheapest star attains equality at its cost. Any other minimizer must therefore have $D = S = 0$. Lemma 11 then gives $L = 0$, and Lemma 12 gives $p = e_v$. Calibration slack zero fixes all remaining code values, so the physical score is exactly F_v. The star costs are distinct, proving uniqueness. □

6 Finite width, the input sequence, and the posterior

Each star has an exact minimum-variation representation obtained by summing signed parity projections. Its number of neurons is at most

$$\sum_{s \in G} 2^{n_s} \le 2^{m + n_{\max}} < N_0.$$

At every $N \ge N_0$ it can therefore be zero-padded and balanced to attain energy $2\sqrt{d}\, C_v$. Conversely (5) and Proposition 13 show that every minimum-energy completion has precisely the cheapest live star score.

Order $\mathcal{V}$ by increasing C_v. The input permutation first lists all physical inputs outside $\mathcal{V}$. Then it lists the complete physical fibers in this cost order, with any fixed order inside each fiber. Designate the first input in each such fiber. Just before this input is queried, the entire preceding fibers and all fibers outside $\mathcal{V}$ have been revealed. The unique minimum-energy score is therefore the star that is positive on this next fiber. The target is negative there. Multiplication by the base character preserves this disagreement.

For completeness, the passage to the actual posterior uses only a finite-dimensional small-noise argument. At a designated round, replace all strict constraints by their closed versions. The feasible set is closed and the quadratic energy is coercive, so minima exist. Requiring the correct sign at the designated point excludes every energy minimizer. It therefore raises the minimum energy by a strictly positive amount. The strict feasible set has the same infimum as its closed version: scale the readout of a minimizing star by $1+\varepsilon$ and let $\varepsilon \downarrow 0$.

Here is a direct Gaussian integral bound. Let E_0 be the minimum energy and $E_1 > E_0$ the minimum on the closed correct-prediction set; if the latter set is empty, its posterior probability is zero. Choose $a \in (E_0, E_1)$. A sufficiently slightly scaled star has a fixed open parameter ball $\mathcal{B}$ that is strictly feasible and satisfies $K_N \le a$ throughout the ball. Put $P = N(d+2)$ and choose $\rho \in (0,1)$ so that $(1-\rho)E_1 > a$. The unnormalized correct-prediction integral is at most

$$\int_{K_N \ge E_1} e^{-K_N/(2\sigma^2)}\, d\theta \le e^{-(1-\rho)E_1/(2\sigma^2)} \int_{\mathbb{R}^P} e^{-\rho K_N/(2\sigma^2)}\, d\theta = c_\rho \sigma^P e^{-(1-\rho)E_1/(2\sigma^2)},$$

where c_ρ is finite and independent of σ. The denominator is at least $\mathrm{vol}(\mathcal{B}) e^{-a/(2\sigma^2)}$. Their ratio tends to zero. The Gaussian normalizing constant and the initial full-margin conditioning cancel from this ratio. This argument sums all parameterizations, including non-minimizing ones; it does not make a tie-breaking assumption.

There are finitely many designated rounds for fixed m. Choose the prior scale small enough that their correct-label probabilities sum to less than η. A union bound then proves the posterior-sampling claim, and probabilities below $1/2$ and $2/3$ prove the other two claims.

It remains to verify the teacher resources. Compute each rail parity, then the m logical parities (each uses at most three rail-pair parities) and their AND; multiplying by the base parity gives f. This uses $O(d)$ bounded-fan-in Boolean gates. For a shallow ReLU teacher use the score

$$F_{\mathrm{teacher}}(x) = \sum_{i=1}^{m} \chi_{A_{e_i}}(x) - (m-1)\chi_{A_0}(x) = b(x)\left(\sum_i (-1)^{z_i} - (m-1)\right). \tag{27}$$

It has margin one and the target sign. Parity on r bits is a piecewise-linear function of their sum on the $r+1$ attainable values. Linear interpolation represents it by at most $r+1$ affine ReLU neurons, with incoming coordinate weights one, biases bounded by r, and readout coefficients bounded by two. The constant term is a neuron on the constant input. Applying this to the $m+1$ parities in (27) gives width at most

$$(n_0+1) + \sum_{i=1}^{m} (n_0 + 5 + 2\deg_{\mathrm{path}}(i)) = (m+1)(n_0+1) + 8m - 4 = C_m.$$

All incoming weights have magnitude at most one, biases have magnitude at most n_0+8, and readout coefficients have magnitude at most $2(m-1)$. Thus every coefficient is bounded by C_m. This completes the proof of Theorem 1.

7 The same lower bound for standard softplus

The conclusion also holds for a smooth activation. Let

$$\phi(t) = \log(1+e^t).$$

Replace ReLU by ϕ in (1), leaving the Gaussian prior, pre-data full-cube margin conditioning, and sign-label likelihood unchanged. In particular the quadratic energy remains (3).

Theorem 14 (Softplus at every sufficiently large width)**.** *Use the target, input permutation, and designated rounds in Theorem 1. For each sufficiently large m there is a finite threshold $N_\phi(m)$ such that, for every fixed $N \geq N_\phi(m)$, the standard softplus learner's correct-label posterior probability tends to zero at every designated round as $\sigma \downarrow 0$. The target has a one-hidden-layer softplus implementation of width at most C_m, output margin greater than one, and every coefficient bounded by $24C_m$. Equivalently, padding to width $\widetilde{C}_m = 24C_m$ gives both width and coefficient bounds $\widetilde{C}_m = \Theta(m^2)$. Thus the cumulative mistake lower bound remains $\exp(\Omega(\widetilde{C}_m^{1/2}))$. The three prediction consequences in Theorem 1 hold for this learner as well.*

The threshold in this extension is existential; the explicit ReLU threshold N_0 need not suffice. The theorem concerns standard softplus, not the bounded activation tanh.

7.1 A bias-independent atom estimate

For a physical weight vector $w \in \mathbb{R}^d$, excluding the constant-input coefficient, define

$$G(w) = \frac{1}{\pi} \sum_s a(n_s) \int_0^\infty \prod_{j \in A_s} |\sin(w_j t)| \prod_{j \notin A_s} |\cos(w_j t)| \, \frac{dt}{t^2}.$$

The coefficient integral (28) implies

$$Q(\mathrm{ReLU}(b + w \cdot x)) \leq G(w) \leq \|w\|_2 \qquad \text{for every } b \in \mathbb{R}. \tag{S1}$$

To justify the second inequality, first sum over all ℓ-pair transversals to obtain G_{full}. On nonnegative block-constant vectors V, this equals the full zero-bias coefficient norm. The generalized Hölder sampling argument in Appendix A therefore gives $G_{\mathrm{full}}(w) \leq \|w\|_2$; retaining only selected nonnegative terms gives $G(w) \leq G_{\mathrm{full}}(w)$.

Let $\mathcal{A}$ be the retained balanced rays in physical weight space. The full-code balanced stability and sampling argument in Appendix A applies to G_{full} without a bias coordinate and gives squared-distance constant $2^9 R^2$. The restriction argument in Lemma 3 then applies to G as well. Indeed, if the closest full ray has omitted transversal A_t, its omitted integral satisfies

$$w_t |\widehat{\mathrm{ReLU}(w \cdot x)}(A_t)| \leq G_t(w) \leq G_{\mathrm{full}}(w) - G(w) \leq \|w\|_2 - G(w).$$

The same Lipschitz coefficient comparison permits using the zero ray in that case. Consequently

$$\mathrm{dist}(w, \mathcal{A})^2 \leq K_c \|w\|_2 (\|w\|_2 - G(w)). \tag{S2}$$

This statement places no restriction on the bias. If $w_* \in \mathcal{A}$, then any function of $b + w_* \cdot x$ belongs to the downward Fourier space $\mathcal{D}$, because it depends only on one selected transversal.

For a centered logistic random variable Z,

$$\phi(t) = \mathbb{E}\,\mathrm{ReLU}(t + Z). \tag{S3}$$

Both sides have derivative $(1 + e^{-t})^{-1}$ and tend to zero as $t \to -\infty$. Averaging (S1) proves $Q(\phi(b + w \cdot x)) \leq G(w) \leq \|w\|_2$. Softplus is 1-Lipschitz, so comparison with a closest $w_* \in \mathcal{A}$ while preserving the bias, together with $\mathbb{E}[xx^\top] = I$, gives

$$\begin{aligned} \|\phi(b + w \cdot x) - P_{\mathcal{D}} \phi(b + w \cdot x)\|_2^2 &\leq K_c \|w\|_2 (\|w\|_2 - G(w)) \\ &\leq K_c \|w\|_2 (\|w\|_2 - Q(\phi(b + w \cdot x))). \end{aligned} \tag{S4}$$

The constant component created by $\phi(0) = \log 2$ lies in $\mathcal{D}$.

7.2 A uniform energy gap

For a softplus network put $B = K_N/(2\sqrt{d})$. The arithmetic–geometric mean inequality gives

$$B \geq \sum_j |v_j| \sqrt{b_j^2 + \|w_j\|_2^2} \geq \sum_j |v_j| \|w_j\|_2.$$

For the stage's code dual $b(x)\lambda(z(x))$, write $D = B - \langle b\lambda, F\rangle$. The atom bound and weighted Cauchy–Schwarz applied to (S4) show

$$Q(F) \leq B, \qquad D \geq 0, \qquad \|F - P_{\mathcal{D}} F\|_2^2 \leq K_c B D. \tag{S5}$$

Since $|\phi(t)| \leq |t| + \log 2$ and $\sum_j |v_j| \leq \sqrt{N \sum_j v_j^2} \leq \sqrt{K_N}$, we also have the width-independent estimate

$$\|F\|_\infty \leq \sqrt{d+1}\, B + (\log 2)\sqrt{2\sqrt{d}\, B}. \tag{S6}$$

Set $B_{\mathrm{cap}} = 5\ell(R+4) \leq 10\ell R$. Every star cost is at most $4.5\ell(R+4)$, so this cap exceeds every star cost by more than one. For $B \leq B_{\mathrm{cap}}$, (S6) is bounded by

$$H_{\mathrm{cap}} = 3\sqrt{d+1}\, B_{\mathrm{cap}}.$$

Repeat the proof of Lemma 11 using (S5) and this uniform supremum bound. It gives

$$L \leq \frac{8 H_{\mathrm{cap}}}{w_0} S + 5 \cdot K_c B_{\mathrm{cap}} D.$$

The two coefficients are at most $480\sqrt{\ell R}$ and $50 \cdot 2^{16} \ell^3 R^5$, respectively, so the same $C_{\mathrm{phys}} = 2^{22} \ell^3 R^5$ applies. The calibrated decomposition, Fourier slack inequalities, and rounding lemma therefore hold with exactly the same B_{round} and absorption estimate.

To make the uniformity in width explicit, consider pairs (B, F) where F is an arbitrary vector of real outputs indexed by the finite input cube. Impose $0 \leq B \leq B_{\mathrm{cap}}$, $\|F\|_\infty \leq H_{\mathrm{cap}}$, the closed full-cube and history margins, and the three inequalities (S5). This is a compact set in a finite-dimensional space. Every softplus network satisfying the closed full-cube and history margins below the energy cap belongs to it, and every ReLU star paired with its exact variation cost also belongs to it.

Only these inequalities are used by the calibrated decomposition, variance control, and rounding argument. Consequently every pair in this compact set satisfies

$$B \geq C_v + (1 - W B_{\mathrm{round}})(D + S)$$

for some live star v. In particular $B \geq C_{\min}$, with equality only for the unique cheapest live star $F_{\min}$. Requiring the correct closed margin at the designated next point excludes that score. Compactness then supplies a strictly positive gap g above $C_{\min}$ on the correct-prediction subset, if that subset is nonempty. Energies above B_{cap} are separated by more than one already. Thus, for some $g' > 0$ independent of learner width, every softplus network satisfying those margins and predicting correctly at this point has

$$K_N \geq 2\sqrt{d}\,(C_{\min} + g'). \tag{S7}$$

7.3 Approaching the star energy at large width

Take a finite ReLU representation of $F_{\min}$ attaining its exact variation cost. Balance its parameters (a_j, c_j) so that

$$d\sum_j \|a_j\|_2^2 + \sum_j c_j^2 = 2\sqrt{d}\, C_{\min}.$$

For every sufficiently large N, use softplus incoming vectors $\sqrt{N}a_j$, readouts $c_j/\sqrt{N}$, and zero readouts on the remaining neurons. Its output is

$$F_N(x) = \sum_j c_j \frac{\phi(\sqrt{N}a_j \cdot \bar{x})}{\sqrt{N}}.$$

Because $0 \le \phi(t) - \mathrm{ReLU}(t) \le \log 2$, these outputs converge uniformly to $F_{\min}$, and their energy before a margin correction is exactly $2\sqrt{d}\, C_{\min}$. Let $\delta_N > 0$ bound the uniform error, with $\delta_N \to 0$. Multiplying the readouts by $1 + 4\delta_N$ gives strict margins for large N, since

$$(1 + 4\delta_N)(1/2 - \delta_N) > 1/2 \qquad (0 < \delta_N < 1/4).$$

The energy still converges to $2\sqrt{d}\, C_{\min}$. It is therefore below the correct-prediction lower bound (S7) for every sufficiently large width. An open parameter ball around this strictly feasible witness and the Gaussian integral comparison from the ReLU proof show that the correct-label posterior probability tends to zero as $\sigma \downarrow 0$ at each such fixed width. There are finitely many designated stages, so one common threshold $N_\phi(m)$ works for all of them.

7.4 A teacher with linearly bounded coefficients

For even r, the parity spline in the ReLU teacher has hinge locations on the grid $-r, -r+2, \dots, r$, hinge coefficients of magnitude at most two, and one constant neuron. At an attainable input sum, at most one hinge argument is zero. The remaining nonzero hinge arguments have absolute values $2, 4, \dots$, with at most two at each distance. Replacing each ReLU term by $\phi(Gt)/G$ therefore changes the parity value by at most

$$\frac{2\log 2 + 4/(e^{2G} - 1) + e^{-G}}{G} < \frac{3}{G} \qquad (G \ge 1).$$

Here the constant neuron contributes at most e^{-G}/G. The logical coefficients in (27) have total absolute value $2m - 1$, so choosing $G = 24m$ makes the target-score error less than $1/4$. Multiply the resulting readout by two. Its margin is then greater than one on every Boolean input.

The hidden width remains C_m. All incoming coordinate weights have magnitude at most $24m$, and biases have magnitude at most $24m(n_0 + 8) \le 24C_m$, since $C_m - m(n_0 + 8) = n_0 + m - 3 > 0$. The final readout coefficients have magnitude at most $4m/G < 1$. Padding to width $24C_m$ with zero readouts preserves the function and puts both width and coefficient bounds under the same parameter. This completes the proof of Theorem 14.

A Proof of the exact unequal-rail norm

Throughout this appendix, m denotes the number of rail pairs in the full-code theorem (called p in Theorem 2, and equal to ℓ in the main construction). We use A_s, Q, and $\mathcal{D}$ for the full transversals, full coefficient norm, and full downward space. Thus the hypothesis is $R \ge 6\Delta$. We also write $n_s = |A_s|$, $w_s = a(n_s)$, $n_{\max} = \max_s n_s$, and $a_{\max} = \max_s a(n_s)$ locally. These conventions do not change the subcode definitions in the main text.

A.1 Two elementary analytic tools

Lemma 15 (Minimum-coordinate parity bound)**.** *For even $n \geq 2$ and any affine ReLU ridge on n Boolean coordinates,*

$$|\widehat{h}([n])| \leq \kappa_n \min_i |w_i|.$$

Proof. A coordinatewise nondecreasing $a : \{-1,1\}^k \to [0,1]$ has top Fourier coefficient at most $2^{-k}\binom{k-1}{\lfloor (k-1)/2 \rfloor}$ in absolute value. To see this, average over coordinate permutations and express the resulting nondecreasing sequence on Hamming levels as a convex combination of level thresholds and a constant. The alternating binomial sum of a threshold starting at level j is $(-1)^{k-j}\binom{k-1}{j-1}$.

Input flips make all incoming weights nonnegative. Integrate out a chosen coordinate i. The difference $[\mathrm{ReLU}(t + w_i) - \mathrm{ReLU}(t - w_i)]/(2w_i)$ lies in $[0,1]$ and is nondecreasing in the other coordinates. Apply the preceding bound with $k = n - 1$. A zero weight gives a zero coefficient directly. □

The affine part of ReLU has no even Fourier coefficient of degree at least two. The identity

$$|z| = \frac{2}{\pi} \int_0^\infty \frac{1 - \cos(tz)}{t^2}\, dt$$

therefore gives, up to its fixed degree sign, the coefficient integral

$$\widehat{h}(A) = \frac{(-1)^{|A|/2+1}}{\pi} \int_0^\infty \cos(bt) \prod_{j \in A} \sin(w_j t) \prod_{j \notin A} \cos(w_j t)\, \frac{dt}{t^2}. \tag{28}$$

For the even block supports under consideration these integrals are absolutely convergent.

Lemma 16 (Block balancing)**.** *For any nonnegative weighted sum of absolute coefficients on nonempty unions of disjoint even blocks, the supremum of its ratio to the augmented incoming Euclidean norm is unchanged by restricting to zero bias and nonnegative weights constant within each block.*

Proof. Take absolute values in (28), discard bias and irrelevant-coordinate cosine factors, and flip signs of incoming weights. Call the resulting nonnegative integral $G(w)/\pi$. For block-constant weights it equals the coefficient sum exactly, because every sine and cosine exponent is even.

For a block of size r, generalized Hölder, on the disjoint union of coefficient-index copies of the integration measure, gives

$$G(w) \leq \prod_{j \text{ in block}} G(w \text{ with this whole block replaced by } w_j)^{1/r}.$$

Iterate over blocks. If V replaces each block by an independent uniformly sampled coordinate magnitude from that block, the result is $G(w)/\pi \leq \exp(\mathbb{E} \log Q(h_V))$. If M_* is the restricted supremum, this is at most $M_* \exp(\mathbb{E} \log \|V\|) \leq M_* \sqrt{\mathbb{E}\|V\|^2} = M_* \|w\|$. Bias and irrelevant weights only increase the original norm. □

A.2 A randomized-minimum comparison for arbitrary rail orientations

Let $m \geq 2$. In pair i, let the dominant-amplitude rail have positive even size p_i and common magnitude a_i, and let the other rail have positive even size q_i and magnitude b_i, where $a_i \geq b_i \geq 0$. Write its independent signs as X_i and Y_i, respectively, and put $D = \sum_i p_i$. Let $h = \mathrm{ReLU}(\sum_i a_i \sum X_i + b_i \sum Y_i)$, and let c_s be the absolute coefficient of the transversal that chooses Y_i precisely when $s_i = 1$.

Lemma 17 (Randomized minimum for unequal even rails)**.** *With independent uniform signs ε_i,*

$$\sum_s c_s \le \kappa_D \, \mathbb{E}_\varepsilon \min_i (a_i + \varepsilon_i b_i).$$

No comparison between p_i and q_i is required.

Proof. The even-block coefficient integral has a fixed sign: $\widehat{h}(A_s) = (-1)^{|A_s|/2+1} c_s$. Put $\eta_i = (-1)^{(q_i-p_i)/2}$ and $\sigma_0 = (-1)^{D/2+1}$. Then

$$\sum_s c_s = \sigma_0 \mathbb{E}\left[\chi_X \prod_i (1 + \eta_i \chi_{X_i} \chi_{Y_i}) h(X,Y)\right] = \sigma_0 \mathbb{E}_X[\chi_X H(X)],$$

where $\chi_X = \prod_i \chi_{X_i}$ and

$$H(X) = \mathbb{E}\left[h(X,Y) \mid \chi_{Y_i} = \eta_i \chi_{X_i} \text{ for every } i\right].$$

Indeed the product is 2^m times the indicator of those independent parity conditions, each of probability $1/2$.

Choose one distinguished coordinate $X_{i,1}, Y_{i,1}$ in each pair. Generate the conditional law by taking independent uniform signs ε_i, setting $Y_{i,1} = \varepsilon_i X_{i,1}$, and choosing the remaining $q_i - 1$ coordinates uniformly subject to

$$\chi_{Y_{i,\mathrm{rest}}} = \eta_i \varepsilon_i \chi_{X_{i,\mathrm{rest}}}.$$

This is precisely the required uniform parity-class law; in particular $H = \mathbb{E}_\varepsilon H_\varepsilon$. Conditional on ε, the distinguished selected coordinate has the nonnegative coefficient $d_i = a_i + \varepsilon_i b_i$. The remaining conditional laws do not depend on any distinguished selected coordinate.

Fix i. If $d_i = 0$, then H_ε is independent of $X_{i,1}$, so its top coefficient is zero. Otherwise its normalized finite derivative

$$G_i(X_{\neq(i,1)}) = \frac{H_\varepsilon(X_{i,1} = 1) - H_\varepsilon(X_{i,1} = -1)}{2d_i}$$

belongs to $[0,1]$ and is coordinatewise nondecreasing. To verify the last assertion, the derivative of a ReLU with distinguished coefficient d_i is the nondecreasing clipped ramp $t \mapsto [\mathrm{ReLU}(t + d_i) - \mathrm{ReLU}(t - d_i)]/(2d_i)$. A different distinguished selected bit changes its argument by $2d_j \ge 0$ without changing any conditional law. Flipping an ordinary selected bit in block j changes the deterministic argument by $2a_j$ and reverses the required parity of $Y_{j,\mathrm{rest}}$. Couple these two uniform parity classes by flipping one fixed remaining opposite bit. Such a bit exists since $q_j \ge 2$. The total argument change is $2a_j - 2b_j Y_{j,k} \ge 0$. This coupling also works when $j = i$, because the conditional law is independent of $X_{i,1}$.

The elementary monotone top-coefficient bound on $D - 1$ bits now gives

$$|\widehat{H}_\varepsilon([D])| = d_i |\widehat{G}_i([D] \setminus \{(i,1)\})| \le d_i \, 2^{-(D-1)} \binom{D-2}{(D-2)/2} = \kappa_D d_i.$$

This holds for every i. Average the bound with the smallest d_i over ε and use the preceding signed identity. □

Lemma 18 (A bound for the mass using one opposite rail)**.** *If $q_i \ge p_i$, then*

$$\sum_{s : s_i = 1} c_s \le \kappa_{D-p_i+q_i} b_i \le \kappa_D b_i.$$

Proof. In the nonnegative coefficient integrals for this sum, discard the cosine factor from the unused selected block X_i. The result is the unweighted coefficient sum for the other rail pairs together with the unpaired selected block Y_i. Repeat the preceding proof for the other pairs and use a distinguished coordinate of Y_i to take the finite derivative. Its coefficient is always b_i; the other conditional couplings are unchanged. The monotone bound therefore gives $\kappa_{D-p_i+q_i} b_i$. Finally κ_n decreases along even integers, by $\kappa_{n+2}/\kappa_n = (n-1)/n$. □

Theorem 19 (Exact balanced norm with a dimension-free rail condition)**.** *Suppose every physical rail size belongs to $[R, R+\Delta]$, where R and Δ are even, $R \geq 2$, and $R \geq 6\Delta$. For the weighted coefficient sum $Q(h) = \sum_s a(|A_s|)c_s$,*

$$Q(h) \leq L := \frac{\sum_i p_i a_i}{\sqrt{D}} \leq \|w\|_2.$$

Let $\mathcal{A}$ be the union of all balanced transversal rays, including zero. If w_* is the orthogonal projection onto the balanced ray on the dominant rails, then

$$\|w - w_*\|_2^2 \leq 2\|w\|_2(\|w\|_2 - Q(h)).$$

Equality in the norm bound requires zero opposite weights and equal dominant coordinate magnitudes.

Proof. Put $t = \mathbb{E}\min_i(a_i + \varepsilon_i b_i)$ and $\delta_i = q_i - p_i$. The independently proved Bernstein representation in Appendix B makes $a(n)$ increasing and concave. Its largest even forward slope is $(a(4) - a(2))/2 = 4 - \sqrt{2} < 3$. Consequently

$$a(|A_s|) \leq a(D) + 3 \sum_{i:\delta_i > 0} \delta_i s_i.$$

The preceding two lemmas imply

$$Q(h) \leq \sqrt{D}\, t + 3\kappa_D \Delta \sum_i b_i.$$

One has $t \leq a_i$. For distinct i, j, concavity of the minimum gives

$$t \leq \mathbb{E}\min(a_i + \varepsilon_i b_i, a_j) = \frac{a_i + a_j - \max(|a_i - a_j|, b_i)}{2},$$

so $b_i \leq (a_i - t) + (a_j - t)$. Summing over ordered pairs yields $\sum_i b_i \leq 2\sum_i (a_i - t)$. Since $\kappa_D\sqrt{D} \leq 1$, $p_i \geq R$, and $R \geq 6\Delta$,

$$Q(h) \leq \sqrt{D}\, t + 6\kappa_D \Delta \sum_i (a_i - t) \leq \sqrt{D}\, t + \frac{1}{\sqrt{D}} \sum_i p_i (a_i - t) = L.$$

The number L is the inner product with the unit balanced selected ray. Thus $\|w - w_*\|_2^2 = \|w\|_2^2 - L^2 \leq 2\|w\|_2(\|w\|_2 - Q(h))$. Equality in $L \leq \|w\|_2$ gives the stated conditions. □

For arbitrary incoming weights, block balancing gives the same norm bound. Equality forces every independently sampled balanced vector to be a balanced selected ray. A partially zero rail, two nonzero opposite rails, or unequal selected coordinate magnitudes would produce a positive-probability strict sample. Bias and irrelevant weights also vanish at equality. Signed parity averaging then proves (9).

A.3 Transferring squared-distance stability before balancing

We now transfer this bound before block balancing without an exponentially small sampling probability. Assume nonnegative relevant weights and independently sample one coordinate magnitude U_i^j from each rail; let V repeat this value throughout the rail. Put $a_i = \max(U_i^0, U_i^1)$, $b_i = \min(U_i^0, U_i^1)$, and $\bar{a} = m^{-1}\sum_i a_i$. The squared scalar distance to selecting one rail of common magnitude is

$$d_0(U)^2 = \sum_i b_i^2 + \sum_i (a_i - \bar{a})^2.$$

Write $\eta = \mathbb{E}d_0(U)^2$ and $r_{\max} = \max r_i^j \le 2R$. For independent rail samples X, Y with $A = \mathbb{E}X^2 \ge B = \mathbb{E}Y^2$ and $A + B > 0$ (the zero case is immediate),

$$\mathbb{E}\min(X^2, Y^2) \ge \frac{\mathbb{E}X^2Y^2}{r_{\max}(A+B)} \ge \frac{B}{2r_{\max}}.$$

Choose deterministically in every pair the rail of larger second moment. Call its sample Z_i and the discarded sample T_i. Then $\mathbb{E}\sum T_i^2 \le 2r_{\max}\eta$ and $|Z_i - a_i| \le T_i$. If P projects off the constant vector in $\mathbb{R}^m$, independence of the Z_i gives

$$\min_\lambda \mathbb{E}\sum_i (Z_i - \lambda)^2 \le \frac{m}{m-1}\mathbb{E}\|PZ\|^2 \le (4 + 8r_{\max})\eta.$$

Adding discarded rails and restoring physical rail-size weights yields

$$\operatorname{dist}(w, \mathcal{A})^2 \le 24 r_{\max}\mathbb{E}\operatorname{dist}(V, \mathcal{A})^2. \tag{29}$$

Normalize the original augmented vector to one and again put $\varepsilon = 1 - Q(h_u)$. For $\varepsilon < 1/2$, the block-balancing geometric-mean inequality and $\mathbb{E}\|V\|^2 \le 1$ imply

$$\mathbb{E}\log\frac{\|V\|}{Q(h_V)} \le -\log(1-\varepsilon) \le 2\varepsilon, \qquad \mathbb{E}\left[1 - \frac{Q(h_V)}{\|V\|}\right] \le 2\varepsilon.$$

Also $\|V\|^2 \le r_{\max}$ pointwise. The balanced squared-distance bound and (29) therefore give

$$\operatorname{dist}(w, \mathcal{A})^2 \le 96 r_{\max}^2 \varepsilon.$$

The bias and irrelevant squared norm are at most 2ε, since $Q(h_u)$ is bounded by the relevant norm. Altogether

$$\operatorname{dist}(u, \mathcal{A})^2 \le 2^9 R^2 \varepsilon.$$

For $\varepsilon \ge 1/2$, the zero ray suffices. Homogeneity proves the corresponding statement at arbitrary norm. Finally ReLU is 1-Lipschitz and $\mathbb{E}[\bar{x}\bar{x}^\top] = I$, so parameter squared distance bounds function squared L^2 distance. Every face atom lies in $\mathcal{D}$, proving (11).

B An exact Bernstein representation

The gamma-function identity for (6) is

$$a(r) = 2\sqrt{\pi r}\frac{\Gamma(r/2)}{\Gamma((r-1)/2)} = 2\sqrt{2\pi}\,A(r/2), \qquad A(z) = \sqrt{z}\frac{\Gamma(z)}{\Gamma(z-1/2)}.$$

Define

$$k(t) = \sqrt{\frac{t}{1-e^{-t}}}, \qquad B(t) = \frac{1}{\pi}\int_0^1 \frac{k(tu)}{\sqrt{u(1-u)}}\,du, \qquad h(t) = \tfrac{1}{2}B'(t) - B''(t).$$

The limits at zero are $k(0) = B(0) = 1$ and $B'(0) = 1/8$. For $t > 0$, let

$$D(t) = \frac{1}{t} - \frac{1}{e^t - 1}.$$

Then $0 < D(t) < 1/2$ and $D'(t) = -1/t^2 + 1/[4\sinh^2(t/2)] < 0$. Since

$$k'/k = D/2, \qquad k''/k = D^2/4 + D'/2 < D/8,$$

we have $B' > 0$ and $B'' < B'/4$. In particular $h > 0$.

Write $\mathcal{L}$ for the Laplace transform and $b_0(t) = (1-e^{-t})^{-1/2} = t^{-1/2}k(t)$. The beta integral and convolution give

$$\frac{\Gamma(z)}{\Gamma(z+1/2)} = \frac{1}{\sqrt{\pi}}\mathcal{L}[b_0](z), \qquad B = \frac{1}{\pi}(b_0 * t^{-1/2}), \qquad z\mathcal{L}[B](z) = 1 + \mathcal{L}[B'](z).$$

Therefore integration by parts yields

$$\begin{aligned} A(z) &= (z-1/2)z\mathcal{L}[B](z) \\ &= z - 3/8 + \mathcal{L}[B'' - B'/2](z) = z - 3/8 - \int_0^\infty e^{-zt}h(t)\,dt. \end{aligned}$$

All functions and derivatives have at most polynomial growth, so these integrals converge for $z > 0$. At $z = 1/2$, A extends continuously to zero. Consequently

$$A(x+1/2) = x + \int_0^\infty e^{-t/2}(1-e^{-xt})h(t)\,dt \quad (x \ge 0),$$

which is the claimed strict Bernstein representation.

For the crude bound $h \le 3$, use $k(t) \le \sqrt{1+t}$ and $D(t) \le \min(1/2, 1/t)$. For $0 < t \le 1$,

$$|D'(t)| = \frac{2(\cosh t - 1) - t^2}{4t^2\sinh^2(t/2)} \le 2(\cosh 1 - 1) - 1 < 1;$$

the numerator's power series starts at degree four and the denominator is at least t^4. For $t \ge 1$, $|D'(t)| \le t^{-2}$. The displayed derivative identities then give $k' \le 1$ and $|k''| \le 2$ on the whole positive line. Hence $B' \le 1$, $|B''| \le 2$, and $0 < h \le 5/2 < 3$. This proves (17) with all bounds used above.